

# 2026년도 전국기능경기대회 채점기준

| 1. 채점상의 유의사항                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 직 종 명 | 클라우드컴퓨팅 |
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| <div>1. 각 모듈의 리소스는 해당 모듈 요구사항에 명시된 Region에 생성한다.</div> <div>2. 웹 페이지 접근은 Chrome 또는 Firefox로 진행한다.</div> <div>3. Shell 명령어의 출력은 AWS CLI 버전에 따라 일부 다를 수 있다.</div> <div>4. 문제지와 채점지에 있는 &lt;비번호&gt;, &lt;자율&gt; 등은 선수 제출 값으로 확인한다.</div> <div>5. 채점은 문항 순서대로 진행한다.</div> <div>6. 부분 점수가 있는 문항은 채점 항목에 표시된 배점만 인정한다.</div> <div>7. 부분 점수가 따로 없는 문항은 조건을 모두 만족해야 점수로 인정한다.</div> <div>8. 리소스 정보를 읽어오는 항목은 CloudShell에서 자동화 채점 스크립트로 조회하고, 채점위원은 출력 결과를 기준으로 판정한다.</div> <div>9. 선수 이의가 있는 경우 채점위원이 동일 명령을 직접 실행하여 확인할 수 있다.</div> <div>10. IAM User Access Key를 별도로 생성하여 채점하지 않는다.</div> <div>11. CloudShell을 실행한 IAM User 또는 Role 권한으로 채점한다.</div> <div>12. 채점 시 대기가 필요한 경우 항목당 최대 3분을 초과하지 않는다.</div> |       |         |

2. 채점기준표

| 1) 주요항목별 배점 |          |                                                    | 직 종 명 |      | 클라우드컴퓨팅 |           |           |     |
|-------------|----------|----------------------------------------------------|-------|------|---------|-----------|-----------|-----|
| 과제<br>번호    | 일련<br>번호 | 주요항목                                               | 배점    | 채점방법 |         | 채점시기      |           | 비고  |
|             |          |                                                    |       | 독립   | 합의      | 경기<br>진행중 | 경기<br>종료후 |     |
| 제2과제        | 1        | DocumentDB based<br>NoSQL Application              | 7.5   | 0    |         |           | 0         | 1모듈 |
|             | 2        | Simplify Service<br>Networking with<br>VPC Lattice | 7.5   | 0    |         |           | 0         | 2모듈 |
|             | 3        | Cloud Event<br>Handling                            | 7.5   | 0    |         |           | 0         | 3모듈 |
|             | 4        | Event-driven Pod<br>Scaling with AWS<br>SQS        | 7.5   | 0    |         |           | 0         | 4모듈 |
| 합계          |          |                                                    | 30    |      |         |           |           |     |

## 2) 채점방법 및 기준

| 과제<br>번호 | 일련<br>번호 | 주요항목                                                     | 일련<br>번호 | 세부항목(채점방법)                                      | 배<br>점   |
|----------|----------|----------------------------------------------------------|----------|-------------------------------------------------|----------|
| 제2과제     | 1        | Document<br>based NoSQL<br>Application                   | 1        | DocumentDB Cluster 및 Instance                   | 1.5      |
|          |          |                                                          | 2        | Secret 및 Client EC2                             | 1.5      |
|          |          |                                                          | 3        | Client Application 및 데이터 적재<br>상태               | 1.5      |
|          |          |                                                          | 4        | DocumentDB Index 및 TTL 구성                       | 1.5      |
|          |          |                                                          | 5        | NoSQL 조회 기능 검증                                  | 1.5      |
|          | 2        | Simplify<br>Service<br>Networking<br>with VPC<br>Lattice | 1        | VPC                                             | 1.5      |
|          |          |                                                          | 2        | Client/Service EC2 및 애플리케이션                     | 1.5      |
|          |          |                                                          | 3        | VPC Lattice Service Network 및<br>Service        | 1.5      |
|          |          |                                                          | 4        | Target Group, Listener, Security<br>Group       | 1.5      |
|          |          |                                                          | 5        | End-to-End 기능 검증                                | 1.5      |
|          | 3        | Cloud Event<br>Handling                                  | 1        | VPC, EC2, Security Group                        | 1.5      |
|          |          |                                                          | 2        | Security Group                                  | 1.5      |
|          |          |                                                          | 3        | SNS Topic 및 Lambda                              | 1.5      |
|          |          |                                                          | 4        | CloudTrail, EventBridge Rule 및<br>Target        | 1.5      |
|          |          |                                                          | 5        | 최종 기능 검증                                        | 1.5      |
|          | 4        | Event-driven<br>Pod Scaling<br>with AWS SQS              | 1        | EKS Cluster, VPC, Fargate Profile               | 1.2<br>5 |
|          |          |                                                          | 2        | SQS Queue, IAM, SA                              | 1.2<br>5 |
|          |          |                                                          | 3        | KEDA/Karpenter                                  | 1.2<br>5 |
|          |          |                                                          | 4        | Worker Application / KEDA<br>ScaledObject       | 1.2<br>5 |
|          |          |                                                          | 5        | Karpenter NodePool, EC2NodeClass,<br>Worker EC2 | 1.2<br>5 |
|          |          |                                                          | 6        | SQS 기반 Scale Out 및 최종 기능<br>검증                  | 1.2<br>5 |

### 3) 채점 내용

| 순번 | 채점 항목                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| 0  | <p>1) CloudShell에서 2과제 채점에 필요한 기본 도구를 확인합니다.<br/> 2) aws, curl, jq, kubectl 명령이 실행 가능한지 확인합니다.<br/> 3) kubectl이 없는 경우 아래 명령어로 EKS Cluster 버전에 맞는 kubectl을 설치 후 다시 확인합니다.<br/> 4) 본 항목은 채점 준비 절차이며 배점은 없습니다.</p> <pre>aws --version curl --version jq --version if command -v kubectl &gt;/dev/null 2&gt;&amp;1; then     kubectl version --client else     EKS_VERSION=\$(aws eks describe-cluster --region us-west-2 --name skills-sqs-cluster --query 'cluster.version' --output text)     KUBECTL_VERSION="v\${EKS_VERSION}.0"     curl -L -o /tmp/kubectl "https://dl.k8s.io/release/\${KUBECTL_VERSION}/bin/linux/amd64/kubectl"     chmod +x /tmp/kubectl     export PATH="/tmp:\$PATH"     kubectl version --client fi</pre> <p>※ kubectl 설치 명령은 CloudShell에 kubectl이 없을 때만 실행합니다.<br/> ※ EKS Cluster가 아직 생성되지 않은 경우 4모듈 채점 직전에 본 설치 절차를 수행합니다.<br/> ※ 채점 스크립트는 kubectl을 자동 설치하지 않으므로, 4모듈 채점 전 본 준비 절차를 완료해야 합니다.</p> <p>5) 아래 명령어를 입력하여 채점에 필요한 값을 변수로 선언합니다.<br/> 6) 값이 비어 있거나 None이면 해당 리소스가 요구 이름으로 생성되지 않은 것입니다.</p> <pre>export NOSQL_CLIENT_EC2_PUBLIC_IP=\$(aws ec2 describe-instances --region ap-northeast-2 --filters Name=tag:Name,Values=skills-nosql-client-ec2 Name=instance-state-name,Values=running --query "Reservations[0].Instances[0].PublicIpAddress" --output text) export LATTICE_CLIENT_EC2_PUBLIC_IP=\$(aws ec2 describe-instances --region</pre> |

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|     | <pre> ap-northeast-1 --filters Name=tag:Name,Values=skills-lattice-client-ec2 Name=instance-state-name,Values=running --query "Reservations[0].Instances[0].PublicIpAddress" --output text) export SERVICE_NETWORK_ID=\$(aws vpc-lattice list-service-networks --region ap-northeast-1 --query "items[?name=='skills-lattice-sn'].id   [0]" --output text) export TARGET_GROUP_ID=\$(aws vpc-lattice list-target-groups --region ap-northeast-1 --query "items[?name=='skills-lattice-order-tg'].id   [0]" --output text) export SERVICE_ID=\$(aws vpc-lattice list-services --region ap-northeast-1 --query "items[?name=='skills-lattice-order-service'].id   [0]" --output text) export SERVICE_EC2_SECURITY_GROUP_ID=\$(aws ec2 describe-instances --region ap-northeast-1 --filters Name=tag:Name,Values=skills-lattice-service-ec2 Name=instance-state-name,Values=running --query "Reservations[0].Instances[0].SecurityGroups[0].GroupId" --output text) export QUEUE_URL=\$(aws sqs get-queue-url --region us-west-2 --queue-name skills-sqs-queue --query QueueUrl --output text) </pre>                                                                                                                                                |
| 1-1 | <p>1) 아래 명령어를 입력합니다.</p> <pre> aws docdb describe-db-clusters --region ap-northeast-2 --db-cluster-identifier skills-nosql-docdb-cluster --query 'DBClusters[0].{Cluster:DBClusterIdentifier,Status:Status,Engine:Engine,Version:En gineVersion,Encrypted:StorageEncrypted,KmsKeyId:KmsKeyId,BackupRetention:BackupRet entionPeriod,Endpoint:Endpoint,Port:Port}' --output table aws docdb describe-db-instances --region ap-northeast-2 --db-instance-identifier skills-nosql-docdb-instance-1 --query 'DBInstances[0].{Instance:DBInstanceIdentifier,Status:DBInstanceStatus,Class:DBIns tanceClass,Engine:Engine,Cluster:DBClusterIdentifier,AZ:AvailabilityZone}' --output table aws kms describe-key --region ap-northeast-2 --key-id alias/skills-nosql-docdb --query 'KeyMetadata.{Arn:Arn,Enabled:Enabled,KeyManager:KeyManager,KeyUsage:KeyUsage}' --output table </pre> <p>2) Cluster=skills-nosql-docdb-cluster, Status=available, Engine=docdb, Encrypted=True, Port=27017, BackupRetention&gt;=1인지 확인합니다.<br/> Instance=skills-nosql-docdb-instance-1, Status=available, Class=db.t3.medium, Cluster=skills-nosql-docdb-cluster인지 확인합니다.<br/> KMS Key는 alias/skills-nosql-docdb로 조회되며 Enabled=True, KeyManager=CUSTOMER인지 확인합니다.</p> |
| 1-2 | <p>1) 아래 명령어를 입력합니다.</p> <pre> aws secretsmanager describe-secret --region ap-northeast-2 --secret-id skills-nosql-docdb-secret --query '{Name:Name,ARN:ARN,KmsKeyId:KmsKeyId}' --output table aws secretsmanager get-secret-value --region ap-northeast-2 --secret-id skills-nosql-docdb-secret --query SecretString --output text   jq -r '{username, </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|     | <p>host, password_set:(.password != null and .password != ""))}'</p> <p>aws ec2 describe-instances --region ap-northeast-2 --filters</p> <p>Name=tag:Name,Values=skills-nosql-client-ec2</p> <p>Name=instance-state-name,Values=running --query</p> <p>'Reservations[].Instances[].{Name:Tags[?Key=`Name`].Value [0],InstanceId:InstanceId,State:State.Name,Type:InstanceType,PublicIp:PublicIpAddress}' --output table</p> <p>2) Secret Name=skills-nosql-docdb-secret이고 SecretString 출력에 username, host, password_set=true가 있어야 합니다.</p> <p>host는 DocumentDB Cluster Endpoint Hostname이어야 하며 Scheme 또는 Port가 포함되면 안 됩니다.</p> <p>Client EC2 Name=skills-nosql-client-ec2, State=running, PublicIp가 존재해야 합니다.</p> |
| 1-3 | <p>1) 아래 명령어를 입력합니다.</p> <pre>curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn" "http://\${NOSQL_CLIENT_EC2_PUBLIC_IP}:8080/health" curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn" "http://\${NOSQL_CLIENT_EC2_PUBLIC_IP}:8080/v1/admin/summary"</pre> <p>2) /health의 http_code=200이고 status=ok, database=skills_retail, port=27017, tls=true가 출력되어야 합니다.</p> <p>/v1/admin/summary의 http_code=200이고 counts.orders&gt;=8, counts.products&gt;=6, counts.sessions&gt;=30이어야 합니다.</p> <p>dateFieldTypes에 orders.createdAt, orders.dueAt, products.updatedAt, sessions.lastSeen, sessions.expiresAt이 날짜 타입으로 표시되어야 합니다.</p>                                                                                                |
| 1-4 | <p>1) 아래 명령어를 입력합니다.</p> <pre>curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn" "http://\${NOSQL_CLIENT_EC2_PUBLIC_IP}:8080/v1/admin/indexes"</pre> <p>2) http_code=200이어야 합니다.</p> <p>orders에는 {orderId:1} unique, {customerId:1, createdAt:-1}, {status:1, dueAt:1} Index가 있어야 합니다.</p> <p>products에는 {productId:1} unique, {warehouseId:1, stock:1} Index가 있어야 합니다.</p> <p>sessions에는 {sessionId:1} unique, {expiresAt:1} TTL expireAfterSeconds=0, {customerId:1, lastSeen:-1} Index가 있어야 합니다.</p> <p>Index 이름은 채점하지 않습니다.</p>                                                                                                                                                                                    |
| 1-5 | <p>1) 아래 명령어를 입력합니다.</p> <pre>curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn" "http://\${NOSQL_CLIENT_EC2_PUBLIC_IP}:8080/v1/orders/0-1001" curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn" "http://\${NOSQL_CLIENT_EC2_PUBLIC_IP}:8080/v1/customers/C001/orders" curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn" "http://\${NOSQL_CLIENT_EC2_PUBLIC_IP}:8080/v1/orders/pending?from=2026-06-01T00:00:00Z&amp;to=2026-06-08T00:00:00Z" curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn" "http://\${NOSQL_CLIENT_EC2_PUBLIC_IP}:8080/v1/products/low-stock?warehouseId=W-A"</pre>                                                                                                                                           |

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|     | <p>2) 모든 조회 API의 http_code=200이어야 합니다.</p> <p>응답 데이터는 retail_dataset.json 기준으로 다음을 만족해야 합니다.</p> <p>0-1001 조회: orderId=0-1001, customerId=C001, status=PENDING, warehouseId=W-A C001 주문 조회: 0-1006, 0-1001, 0-1004 포함</p> <p>기간 내 PENDING 조회: 0-1001, 0-1003, 0-1006 포함</p> <p>W-A Low Stock 조회: P-BLU-003, P-RED-001 포함 및 P-GRN-002 미포함</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2-1 | <p>1) 아래 명령어를 입력합니다.</p> <pre>aws ec2 describe-vpcs --region ap-northeast-1 --filters Name=tag:Name,Values=skills-lattice-client-vpc,skills-lattice-service-vpc --query 'Vpcs[].{Name:Tags[?Key=='Name'].Value [0],VpcId:VpcId,Cidr:CidrBlock,State:State} ' --output table</pre> <p>2) skills-lattice-client-vpc의 Cidr=10.61.0.0/16, State=available이어야 합니다.<br/>skills-lattice-service-vpc의 Cidr=10.62.0.0/16, State=available이어야 합니다.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2-2 | <p>1) 아래 명령어를 입력합니다.</p> <pre>aws ec2 describe-instances --region ap-northeast-1 --filters Name=tag:Name,Values=skills-lattice-client-ec2,skills-lattice-service-ec2 Name=instance-state-name,Values=running --query 'Reservations[].Instances[].{Name:Tags[?Key=='Name'].Value [0],Id:InstanceId,Type: InstanceType,PublicIp:PublicIpAddress,PrivateIp:PrivateIpAddress,State:State.Name} ' --output table</pre> <p>LATTICE_CLIENT_EC2_PUBLIC_IP=\$(aws ec2 describe-instances --region ap-northeast-1<br/>--filters Name=tag:Name,Values=skills-lattice-client-ec2<br/>Name=instance-state-name,Values=running --query<br/>'Reservations[0].Instances[0].PublicIpAddress' --output text 2&gt;/dev/null    true)<br/>curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn"<br/>"http://\${LATTICE_CLIENT_EC2_PUBLIC_IP}/health"</p> <p>2) skills-lattice-client-ec2와 skills-lattice-service-ec2가 State=running이어야 합니다.</p> <p>skills-lattice-client-ec2는 PublicIp가 존재해야 하고, skills-lattice-service-ec2는 PublicIp가 없어야 합니다.</p> <p>Client /health 응답의 http_code=200이고 status=ok, app=client가 포함되어야 합니다.</p> |
| 2-3 | <p>1) 아래 명령어를 입력합니다.</p> <pre>SERVICE_NETWORK_ID=\$(aws vpc-lattice list-service-networks --region ap-northeast-1 --query 'items[?name=='skills-lattice-sn'].id [0]' --output text 2&gt;/dev/null    true)</pre> <pre>aws vpc-lattice list-service-networks --region ap-northeast-1 --query 'items[?name=='skills-lattice-sn'].{Name:name,Id:id,AssociatedVPCs:numberOfAssocia tedVPCs,AssociatedServices:numberOfAssociatedServices}' --output table</pre> <pre>aws vpc-lattice list-services --region ap-northeast-1 --query 'items[?name=='skills-lattice-order-service'].{Name:name,Id:id,Dns:dnsEntry.domain Name,Status:status}' --output table</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|     | <pre>aws vpc-lattice list-service-network-vpc-associations --region ap-northeast-1 --service-network-identifier "\$SERVICE_NETWORK_ID" --query 'items[].{AssociationId:id,VpcId:vpclId,Status:status}' --output table VPC_ASSOCIATION_ID=\$(aws vpc-lattice list-service-network-vpc-associations --region ap-northeast-1 --service-network-identifier "\$SERVICE_NETWORK_ID" --query 'items[0].id' --output text 2&gt;/dev/null    true) echo "VPC_ASSOCIATION_ID=\${VPC_ASSOCIATION_ID}" aws vpc-lattice get-service-network-vpc-association --region ap-northeast-1 --service-network-vpc-association-identifier "\$VPC_ASSOCIATION_ID" --query '{AssociationId:id,VpcId:vpclId,Status:status,SecurityGroupIds:securityGroupIds}' --output table aws vpc-lattice list-service-network-service-associations --region ap-northeast-1 --service-network-identifier "\$SERVICE_NETWORK_ID" --query 'items[].{ServiceId:serviceId,Status:status,Dns:dnsEntry.domainName}' --output table</pre> <p>2) Service Network Name=skills-lattice-sn이 존재해야 합니다.<br/> Service Name=skills-lattice-order-service가 존재하고 Status=ACTIVE이며 Dns가<br/> 출력되어야 합니다.<br/> Client VPC Association의 Status=ACTIVE이어야 합니다.<br/> Service Association의 Status=ACTIVE이어야 합니다.</p>                                                                                                    |
| 2-4 | <p>1) 아래 명령어를 입력합니다.</p> <pre>TARGET_GROUP_ID=\$(aws vpc-lattice list-target-groups --region ap-northeast-1 --query 'items[?name=='skills-lattice-order-tg'].id [0]' --output text 2&gt;/dev/null    true) SERVICE_ID=\$(aws vpc-lattice list-services --region ap-northeast-1 --query 'items[?name=='skills-lattice-order-service'].id [0]' --output text 2&gt;/dev/null    true) SERVICE_SG_IDS=\$(aws ec2 describe-instances --region ap-northeast-1 --filters Name=tag:Name,Values=skills-lattice-service-ec2 Name=instance-state-name,Values=running --query 'Reservations[0].Instances[0].SecurityGroups[].GroupId' --output text 2&gt;/dev/null    true) aws vpc-lattice list-target-groups --region ap-northeast-1 --query 'items[?name=='skills-lattice-order-tg'].{Name:name,Id:id,Type:type,Port:port,Prot ocol:protocol,Vpc:vpclIdentifier,Status:status}' --output table aws vpc-lattice list-targets --region ap-northeast-1 --target-group-identifier "\$TARGET_GROUP_ID" --query 'items[].{Target:id,Port:port,Status:status}' --output table aws vpc-lattice list-listeners --region ap-northeast-1 --service-identifier "\$SERVICE_ID" --query 'items[?name=='skills-lattice-http-listener'].{Name:name,Id:id,Port:port,Protocol: protocol}' --output table aws ec2 describe-security-groups --region ap-northeast-1 --group-ids</pre> |



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|     | <p><code>\$SERVICE_SG_IDS --query</code><br/> <code>'SecurityGroups[].{GroupId:GroupId,GroupName:GroupName,VpcId:VpcId,Inbound:IpPermissions}' --output json</code></p> <p>2) Target Group Name=skills-lattice-order-tg, Type=INSTANCE, Protocol=HTTP, Port=8080이어야 합니다.</p> <p>Target은 skills-lattice-service-ec2 Instance이고 Status=HEALTHY이어야 합니다.</p> <p>Listener Name=skills-lattice-http-listener, Protocol=HTTP, Port=80이어야 합니다.</p> <p>Service EC2 Security Group의 TCP/8080 Inbound는 VPC Lattice Managed Prefix List로 제한되어야 하며 0.0.0.0/0 허용 시 미충족입니다.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2-5 | <p>1) 아래 명령어를 입력합니다.</p> <p><code>LATTICE_CLIENT_EC2_PUBLIC_IP=\$(aws ec2 describe-instances --region ap-northeast-1 --filters Name=tag:Name,Values=skills-lattice-client-ec2 Name=instance-state-name,Values=running --query</code><br/> <code>'Reservations[0].Instances[0].PublicIpAddress' --output text 2&gt;/dev/null    true)</code><br/> <code>curl -s -m 10 -w "Wnhttp_code=%{http_code}Wn"</code><br/> <code>"http://\${LATTICE_CLIENT_EC2_PUBLIC_IP}/v1/client/orders?id=1001 "</code></p> <p>2) http_code=200이어야 합니다.</p> <p>응답 JSON에 client=ok, service.order_id=1001, service.via=vpc-lattice가 포함되어야 합니다.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3-1 | <p>1) 아래 명령어를 입력합니다.</p> <p><code>aws ec2 describe-vpcs --region ap-southeast-1 --filters</code><br/> <code>Name=tag:Name,Values=skills-ceh-vpc --query</code><br/> <code>'Vpcs[].{Name:Tags[?Key=`Name`].Value [0],VpcId:VpcId,Cidr:CidrBlock}' --output table</code></p> <p><code>aws ec2 describe-instances --region ap-southeast-1 --filters</code><br/> <code>Name=tag:Name,Values=skills-ceh-ec2 Name=instance-state-name,Values=running --query</code><br/> <code>'Reservations[] .Instances[] .{Name:Tags[?Key=`Name`].Value [0],Id:InstanceId,Type: InstanceType,State:State.Name,SecurityGroups:SecurityGroups[].GroupId}' --output table</code></p> <p><code>aws ec2 describe-security-groups --region ap-southeast-1 --filters</code><br/> <code>Name=tag:Name,Values=skills-ceh-protected-sg --query</code><br/> <code>'SecurityGroups[].{Name:Tags[?Key=`Name`].Value [0],GroupId:GroupId,GroupName:Group Name,VpcId:VpcId}' --output table</code></p> <p>2) VPC Name=skills-ceh-vpc, Cidr=10.73.0.0/16이어야 합니다.</p> <p>EC2 Name=skills-ceh-ec2, State=running이어야 합니다.</p> <p>Security Group Name=skills-ceh-protected-sg가 존재하고 EC2에 연결되어 있어야 합니다.</p> |
| 3-2 | <p>1) 아래 명령어를 입력합니다.</p> <p><code>aws ec2 describe-security-groups --region ap-southeast-1 --filters</code><br/> <code>Name=tag:Name,Values=skills-ceh-protected-sg --query</code><br/> <code>'SecurityGroups[].{GroupId:GroupId,Inbound:IpPermissions,Outbound:IpPermissionsEgr</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|     | <p>ess}' --output json</p> <p>2) Inbound 또는 IpPermissions가 빈 배열 []이어야 합니다.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3-3 | <p>1) 아래 명령어를 입력합니다.</p> <pre>aws sns list-topics --region ap-southeast-1 --query 'Topics[?contains(TopicArn, `:skills-ceh-alert-topic`)].TopicArn' --output table</pre> <pre>aws lambda get-function-configuration --region ap-southeast-1 --function-name skills-ceh-remediate-fn --query '{FunctionName:FunctionName,State:State,LastUpdateStatus:LastUpdateStatus,Runtime:Runtime,Handler:Handler,Timeout:Timeout,Role:Role,Environment:Environment.Variables}' --output table</pre> <p>2) SNS Topic ARN에 skills-ceh-alert-topic이 포함되어야 합니다.</p> <p>Lambda FunctionName=skills-ceh-remediate-fn, State=Active, Runtime=python3.12, Handler=remediate_security_group.lambda_handler, Timeout&gt;=30이어야 합니다.</p> <p>Environment에는 PROTECTED_SECURITY_GROUP_ID와 SNS_TOPIC_ARN이 존재해야 합니다.</p>                                                                                                                                                                                                                                                                                                                              |
| 3-4 | <p>1) 아래 명령어를 입력합니다.</p> <pre>aws cloudtrail get-trail-status --region ap-southeast-1 --name skills-ceh-cloudtrail --query '{IsLogging:IsLogging,LatestDeliveryTime:LatestDeliveryTime,LatestDeliveryError:LatestDeliveryError}' --output table</pre> <pre>aws events describe-rule --region ap-southeast-1 --name skills-ceh-sg-change-rule --event-bus-name default --query '{Name:Name,State:State,EventPattern:EventPattern}' --output json</pre> <pre>aws events list-targets-by-rule --region ap-southeast-1 --rule skills-ceh-sg-change-rule --event-bus-name default --query 'Targets[0].{Id:Id,Arn:Arn}' --output table</pre> <pre>aws lambda get-policy --region ap-southeast-1 --function-name skills-ceh-remediate-fn --query 'Policy' --output text</pre> <p>2) CloudTrail IsLogging=True이어야 합니다.</p> <p>EventBridge Rule Name=skills-ceh-sg-change-rule, State=ENABLED이어야 합니다.</p> <p>EventPattern에 AuthorizeSecurityGroupIngress와 EC2 API Call via CloudTrail 조건이 포함되어야 합니다.</p> <p>Target Arn은 skills-ceh-remediate-fn Lambda ARN이어야 합니다.</p> <p>Lambda Policy에 events.amazonaws.com 호출 권한이 포함되어야 합니다.</p> |
| 3-5 | <p>1) 아래 명령어를 입력합니다.</p> <pre>export PROTECTED_SECURITY_GROUP_ID=\$(aws ec2 describe-security-groups --region ap-southeast-1 --filters Name=tag:Name,Values=skills-ceh-protected-sg --query 'SecurityGroups[0].GroupId' --output text)</pre> <pre>aws ec2 authorize-security-group-ingress --region ap-southeast-1 --group-id "\$PROTECTED_SECURITY_GROUP_ID" --protocol tcp --port 22 --cidr 0.0.0.0/0</pre> <pre>jq -n --arg sg "\$PROTECTED_SECURITY_GROUP_ID"</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|     | <pre>'{detail:{eventName:"AuthorizeSecurityGroupIngress",requestParameters:{groupId:\$sg }}}' &gt; /tmp/skills-ceh-remediate-event.json aws lambda invoke --region ap-southeast-1 --function-name skills-ceh-remediate-fn --cli-binary-format raw-in-base64-out --payload file:///tmp/skills-ceh-remediate-event.json /tmp/skills-ceh-remediate-output.json aws ec2 describe-security-groups --region ap-southeast-1 --group-ids "\$PROTECTED_SECURITY_GROUP_ID" --query 'SecurityGroups[0].IpPermissions' --output json aws logs describe-log-groups --region ap-southeast-1 --log-group-name-prefix /aws/lambda/skills-ceh-remediate-fn --query 'logGroups[].logGroupName' --output table</pre> <p>2) Lambda Invoke가 성공해야 합니다.<br/> 180초 이내 SecurityGroups[0].IpPermissions가 빈 배열 []이어야 합니다.<br/> /aws/lambda/skills-ceh-remediate-fn Log Group이 출력되어야 합니다.<br/> 본 항목은 CloudTrail 전달 지연 시간을 채점하지 않습니다.</p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| 4-1 | <p>1) 아래 명령어를 입력합니다.</p> <pre>aws eks describe-cluster --region us-west-2 --name skills-sqs-cluster --query 'cluster.{Name:name,Status:status,Endpoint:endpoint,VpcId:resourcesVpcConfig.vpcId ,Subnets:resourcesVpcConfig.subnetIds}' --output table aws eks describe-fargate-profile --region us-west-2 --cluster-name skills-sqs-cluster --fargate-profile-name skills-sqs-fp-keda --query 'fargateProfile.{Name:fargateProfileName,Status:status,Namespaces:selectors[].name space}' --output table aws eks describe-fargate-profile --region us-west-2 --cluster-name skills-sqs-cluster --fargate-profile-name skills-sqs-fp-karpenter --query 'fargateProfile.{Name:fargateProfileName,Status:status,Namespaces:selectors[].name space}' --output table aws eks update-kubeconfig --region us-west-2 --name skills-sqs-cluster kubectl get nodes -l eks.amazonaws.com/compute-type=fargate -o json   jq ' [.items[]   {name:.metadata.name}]'</pre> <p>2) Cluster Name=skills-sqs-cluster, Status=ACTIVE, Endpoint 출력, VpcId 및 Subnets 출력이 있어야 합니다.<br/> Fargate Profile skills-sqs-fp-keda와 skills-sqs-fp-karpenter가 Status=ACTIVE이어야 합니다.<br/> skills-sqs-fp-keda Namespaces에 keda, skills-sqs-fp-karpenter Namespaces에 karpenter가 포함되어야 합니다.<br/> CloudShell에서 kubectl 접근이 가능하고 Fargate Node 이름이 출력되어야 합니다.</p> |
| 4-2 | <p>1) 아래 명령어를 입력합니다.</p> <pre>QUEUE_URL=\$(aws sqs get-queue-url --region us-west-2 --queue-name skills-sqs-queue --query QueueUrl --output text 2&gt;/dev/null    true) echo "QUEUE_URL=\${QUEUE_URL}" if [ -n "\$QUEUE_URL" ] &amp;&amp; [ "\$QUEUE_URL" != "None" ]; then</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|     | <pre>aws sqs get-queue-attributes --region us-west-2 --queue-url "\$QUEUE_URL" --attribute-names QueueArn VisibilityTimeout --output table else     echo "skills-sqs-queue Queue URL 식별 실패" fi kubectl get serviceaccount keda-operator -n keda -o jsonpath='keda/keda-operator role={.metadata.annotations.eksW.amazonawsW.com/role-arn}{ "Wn"}' kubectl get serviceaccount karpenter -n karpenter -o jsonpath='karpenter/karpenter role={.metadata.annotations.eksW.amazonawsW.com/role-arn}{ "Wn"}' kubectl get serviceaccount sqs-worker-sa -n skills-sqs -o jsonpath='skills-sqs/sqs-worker-sa role={.metadata.annotations.eksW.amazonawsW.com/role-arn}{ "Wn"}'</pre> <p>2) Queue URL이 출력되어야 합니다.<br/>QueueArn이 출력되고 VisibilityTimeout&gt;=300이어야 합니다.<br/>keda/keda-operator, karpenter/karpenter, skills-sqs/sqs-worker-sa의 role-arn<br/>annotation이 비어 있지 않아야 합니다.</p>                                                                                                                                |
| 4-3 | <p>1) 아래 명령어를 입력합니다.</p> <pre>kubectl get deployment keda-operator -n keda -o json   jq '{name:.metadata.name, availableReplicas:(.status.availableReplicas // 0), readyReplicas:(.status.readyReplicas // 0)}'</pre> <pre>kubectl get pods -n keda -o json   jq ' [.items[]   select(.metadata.name   test("^keda-operator-"))   {name:.metadata.name, phase:.status.phase, node:.spec.nodeName}]'</pre> <pre>kubectl get deployment karpenter -n karpenter -o json   jq '{name:.metadata.name, availableReplicas:(.status.availableReplicas // 0), readyReplicas:(.status.readyReplicas // 0)}'</pre> <pre>kubectl get pods -n karpenter -o json   jq ' [.items[]   select(.metadata.name   test("^karpenter-"))   {name:.metadata.name, phase:.status.phase, node:.spec.nodeName}]'</pre> <p>2) keda-operator Deployment의 availableReplicas가 1 이상이어야 합니다.<br/>karpenter Deployment의 availableReplicas가 1 이상이어야 합니다.<br/>각 Deployment Pod의 phase는 Running이어야 합니다.<br/>해당 Pod들은 Fargate Node에서 실행되어야 합니다.</p> |
| 4-4 | <p>1) 아래 명령어를 입력합니다.</p> <pre>kubectl get deployment sqs-worker -n skills-sqs -o jsonpath='name={.metadata.name}{ "Wn"}serviceAccountName={.spec.template.spec.servi ceAccountName}{ "Wn"}selector={.spec.selector.matchLabels}{ "Wn"}podLabels={.spec.te mplate.metadata.labels}{ "Wn"}nodeSelector={.spec.template.spec.nodeSelector}{ "Wn"} env={.spec.template.spec.containers[0].env}{ "Wn"}image={.spec.template.spec.contai ners[0].image}{ "Wn"}'</pre> <pre>kubectl get scaledobject sqs-worker-scaledobject -n skills-sqs -o json   jq '{name:.metadata.name, namespace:.metadata.namespace,</pre>                                                                                                                                                                                                                                                                                                                                                                                                   |

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|     | <p>scaleTargetRef:.spec.scaleTargetRef, minReplicaCount:.spec.minReplicaCount, maxReplicaCount:.spec.maxReplicaCount, pollingInterval:.spec.pollingInterval, cooldownPeriod:.spec.cooldownPeriod, triggers:.spec.triggers}'</p> <p>kubectl get triggerauthentication sqs-worker-trigger-auth -n skills-sqs -o json   jq '{name:.metadata.name, namespace:.metadata.namespace, podIdentity:.spec.podIdentity}'</p> <p>2) Deployment name=sqs-worker, serviceAccountName=sqs-worker-sa이어야 합니다. selector/podLabels에 app=sqs-worker가 포함되어야 합니다. nodeSelector에 karpenter.sh/nodepool=sqs-sqs-nodepool, skills-nodepool=event-worker가 포함되어야 합니다. env에 SQS_QUEUE_URL, AWS_REGION, PROCESSING_SECONDS가 있어야 합니다. ScaledObject name=sqs-worker-scaledobject, scaleTargetRef.name=sqs-worker, minReplicaCount=0, maxReplicaCount=6, pollingInterval&lt;=15, cooldownPeriod&lt;=30이어야 합니다. Trigger type은 aws-sqs-queue이고 queueLength=20이어야 합니다. TriggerAuthentication name=sqs-worker-trigger-auth, namespace=skills-sqs가 출력되어야 합니다. podIdentity.provider=aws-eks이어야 합니다.</p>                                                                                                                                                                                                                                                                                                                                                  |
| 4-5 | <p>1) 아래 명령어를 입력합니다.</p> <p>kubectl get nodepool skills-sqs-nodepool -o json   jq '{name:.metadata.name, labels:.spec.template.metadata.labels, nodeClassRef:.spec.template.spec.nodeClassRef, requirements:.spec.template.spec.requirements, consolidationPolicy:.spec.disruption.consolidationPolicy}'</p> <p>kubectl get ec2nodeclass skills-sqs-nodeclass -o json   jq '{name:.metadata.name, role:.spec.role, instanceProfile:.spec.instanceProfile, subnetSelectorTerms:.spec.subnetSelectorTerms, securityGroupSelectorTerms:.spec.securityGroupSelectorTerms, amiFamily:.spec.amiFamily}'</p> <p>kubectl get nodes -l karpenter.sh/nodepool=sqs-sqs-nodepool,skills-nodepool=event-worker -o json   jq '[.items[]   {name:.metadata.name, nodepool:.metadata.labels["karpenter.sh/nodepool"], skillsNodepool:.metadata.labels["skills-nodepool"], instanceType:.metadata.labels["node.kubernetes.io/instance-type"], ready:(.status.conditions[]   select(.type=="Ready"))[0].status}]'</p> <p>kubectl get pods -n skills-sqs -l app=sqs-worker -o json   jq '[.items[]   {name:.metadata.name, phase:.status.phase, node:.spec.nodeName}]'</p> <p>2) NodePool name=skills-sqs-nodepool이어야 합니다. NodePool name=skills-sqs-nodepool이어야 합니다. NodePool labels에 skills-nodepool=event-worker가 있어야 합니다. NodePool nodeClassRef.name=skills-sqs-nodeclass이어야 합니다. NodePool consolidationPolicy가 비어 있지 않아야 합니다.</p> |

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|     | <p>EC2NodeClass name=skills-sqs-nodeclass이어야 하며 role 또는 instanceProfile이 출력되어야 합니다.</p> <p>Worker EC2 Node는 nodepool=skills-sqs-nodepool, skillsNodepool=event-worker, ready=True가 출력되어야 합니다.</p> <p>Worker Pod는 phase와 node가 출력되어야 합니다.</p> <p>단, Worker Deployment 및 KEDA 설정의 minReplicaCount=0 특성상 채점 시점에 Worker EC2 Node 또는 Worker Pod가 없을 수 있으므로, 이 경우 동일 과제의 4-6 SQS 기반 Scale Out 실행 결과에서 확인된 Worker EC2 Node 및 Worker Pod 출력을 포함하여 판정할 수 있습니다.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4-6 | <p>1) 아래 명령어로 SQS 메시지를 12개 발행합니다.</p> <pre> if [ -z "\$QUEUE_URL" ]    [ "\$QUEUE_URL" = "None" ]; then     echo "skills-sqs-queue Queue URL 식별 실패" else     SENT=0     for I in \$(seq 1 12); do         aws sqs send-message --region us-west-2 --queue-url "\$QUEUE_URL" --message-body "judge-\$I" &gt;/dev/null 2&gt;&amp;1 &amp;&amp; SENT=\$((SENT + 1))     done     echo "sent=\${SENT}"     for T in 60 120 180; do         sleep 60         echo "after_\${T}s"         aws sqs get-queue-attributes --region us-west-2 --queue-url "\$QUEUE_URL" --attribute-names ApproximateNumberOfMessages ApproximateNumberOfMessagesNotVisible ApproximateNumberOfMessagesDelayed --output table         kubectl get deployment sqs-worker -n skills-sqs         kubectl get pods -n skills-sqs -l app=sqs-worker -o wide         kubectl get nodes -l         karpenter.sh/nodepool=skills-sqs-nodepool,skills-nodepool=event-worker -o wide         kubectl get nodeclaims -l karpenter.sh/nodepool=skills-sqs-nodepool     done fi </pre> <p>2) SQS 메시지 발행 수가 12개이어야 합니다.</p> <p>마지막 메시지 발행 완료 후 180초 이내 sqs-worker Pod가 증가해야 합니다.</p> <p>180초 이내 Karpenter EC2 Worker Node가 1개 이상 Ready 상태로 확인되어야 합니다.</p> <p>SQS ApproximateNumberOfMessages가 감소해야 합니다.</p> <p>ApproximateNumberOfMessagesNotVisible은 Worker가 처리 중인 메시지이므로, 대기 메시지가 감소하고 Pod가 처리 중이면 실패로 보지 않습니다.</p> <p>Scale In 완료 시간은 채점하지 않습니다.</p> <p>본 항목은 SQS 메시지를 생성하므로 4모듈의 마지막 채점 항목으로 수행합니다.</p> |

